

T05 RADIANT THERMAL COMFORT | O (MAX : 2 PT)

Intent : Maximize volume of the space, reduce dust transmission, improve ventilation control and increase thermal comfort by incorporating radiant heat and cooling systems into the building design.

Summary : This WELL feature requires projects to use radiant systems and independently controlled ventilation systems.

Issue : Mean radiant temperature is one of the six core thermal comfort parameters. It is influenced by a surface material's ability to absorb or emit radiant heat, the extent to which the surface area is exposed to the person (view factor) and the temperatures of the surrounding objects. Uniform thermal radiation may be disrupted by cold windows, uninsulated walls, equipment and improperly sized heating panels, causing local thermal discomfort.¹ In addition, conventional "all-air" systems have a higher risk of draft discomfort, due to elevated indoor air velocity.² Studies show occupants have a 50% probability of indoor temperature satisfaction in spaces with radiant systems, compared to conventional "all-air" systems.³

Solutions : Radiant heating systems are designed to affect mean radiant temperature, and thus the heat exchange with the people in the space, by supplying heat directly to the surrounding surfaces of the floors, walls and ceilings. With radiant systems, the heat transfer due to radiation increases, while the heat transfer due to convection decreases. Buildings with radiant systems have the advantage of quiet operation, low energy consumption and the capability of design integration with an independent ventilation system.⁴ Coupling a radiant system with a dedicated air system separates the twofold role of the mechanical system controlling both heating/cooling and ventilation allowing for better control of the thermal environment. The use of radiant heating and cooling also reduces the number of allergens circulated in the air, since this type of system does not use forced air to distribute heating or cooling.

PART 1 IMPLEMENT RADIANT HEATING (MAX : 1 PT)

For All Spaces:

At least 50% of the regularly occupied project area is heated with one or more of the following:

- a. Radiant ceilings, walls or floors.
- b. Radiant panels which cover at least half of the wall or ceiling to which they are attached (does not include steam radiators).

PART 2 IMPLEMENT RADIANT COOLING (MAX : 1 PT)

For All Spaces:

At least 50% of the regularly occupied project area is cooled with one or more of the following:

- a. Radiant ceilings, walls or floors.
- b. Radiant panels that cover at least half of the wall or ceiling to which they are attached.

REFERENCES

1. American Society of Heating Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook: fundamentals. 2017. <https://www.ashrae.org/technical-resources/ashrae-handbook/description-2017-ashrae-handbook-fundamentals>.
2. Karmann C, Schiavon S, Bauman F. Thermal comfort in buildings using radiant vs. all-air systems: A critical literature review. Build Environ. 2017;111:123-131. doi:10.1016/j.buildenv.2016.10.020
3. Karmann C, Schiavon S, Graham LT, Raftery P, Bauman F. Comparing temperature and acoustic satisfaction in 60 radiant and all-air buildings. Build Environ. 2017;126(January 2018):431-441. doi:10.1016/j.buildenv.2017.10.024
4. Rhee KN, Olesen BW, Kim KW. Ten questions about radiant heating and cooling systems. Build Environ. 2017;112:367-381. doi:10.1016/j.buildenv.2016.11.030